

Report the total number of inodes and the percentage being used in your file system.

- Filesystem: /dev/sda3
- Total Inodes: 5,210,112
- Inodes Used: 289,980
- Inodes Free: 4,920,132
- Inode Usage Percentage: 6%

Current Blocksize for System

- 4096 bytes

Inode Number for the Created File:

- 824775

File Information:

- Type: Regular
- Mode (Permissions): 0664
- Flags: 0x80000
- Links: 1
- Inode checksum: 0x6c844d17
- Extents: 6906997

Report Detailed File Information (Soft Link):

- Inode for file.txt: 824775
- Inode for hi.txt (soft link): 824776
- File Modes in ASCII: file.txt (-rw-rw-r--)
hi.txt (lrwxrwxrwx)
- File Sizes: file.txt: 56 bytes
hi.txt: 8 bytes
- File Names: file.txt listed as file.txt and hi.txt -> file.txt

Report Detailed File Information (Hard Link):

- Inode for file1.txt: 824777
- Inode for bye.txt: 824777
- File Modes in ASCII: file1.txt (-rw-rw-r--)
bye.txt (-rw-rw-r--)
- File Sizes: file1.txt: 77 bytes
bye.txt: 77 bytes
- File Names: file1.txt listed as file1.txt and bye.txt listed as bye.txt

Difference between SoftLink and HardLink:

- Soft Link:
 - Acts as a reference or shortcut to the file's name.
 - Possesses a unique inode distinct from the file it references.
 - If its target file is removed, the link is rendered non-functional or "orphaned."

- Hard Link:
 - Connects directly to the file's actual storage location on the disk, known as an inode.
 - Does not maintain a separate inode; instead, it shares the inode with the file to which it links.
 - Continues to be operative and retains access to the content, even if the file's original name is removed, provided at least one hard link to the file's inode remains.